Algorithmic State Machine (ASM) Charts

- For larger machines the designers often use a different form of representation, called the *algorithmic state machine (ASM) chart*.

- An ASM chart is a type of flowchart that can be used to represent the state transitions and generated outputs for an FSM.

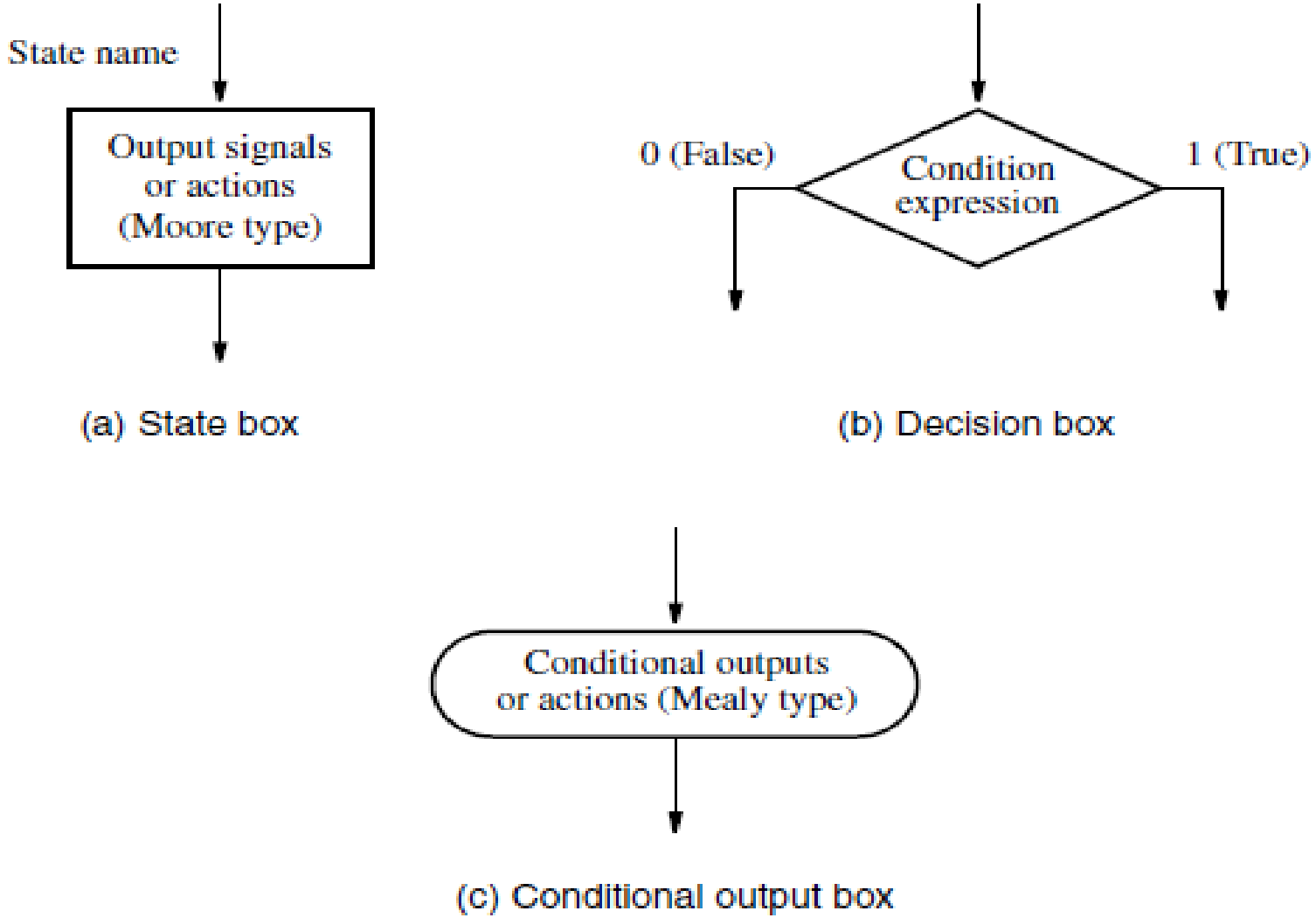


Figure 6.81 Elements used in ASM charts.

State box:

- A rectangle represents a state of the FSM. It is equivalent to a node in the state diagram or a row in the state table. The name of the state is indicated outside the box in the top-left corner.
- The Moore-type outputs are listed inside the box. It is customary to write only the name of the signal that has to be asserted. Thus it is sufficient to write z, rather than $z = 1$, to indicate that the output z must have the value 1.
- Also, it may be useful to indicate an action that must be taken; for example, $Count \leftarrow Count + 1$ specifies that the contents of a counter have to be incremented by 1.

Decision box:

- A diamond indicates that the stated condition expression is to be tested and the exit path is to be chosen accordingly.
- The condition expression consists of one or more inputs to the FSM. For example, *w* indicates that the decision is based on the value of the input *w*, whereas *w*1 ▪ *w*2 indicates that the true path is taken if *w*1 = *w*2 = 1 and the false path is taken otherwise.

Conditional output box:

- An oval denotes the output signals that are of Mealy type.

- The condition that determines whether such outputs are generated is specified in a decision box.

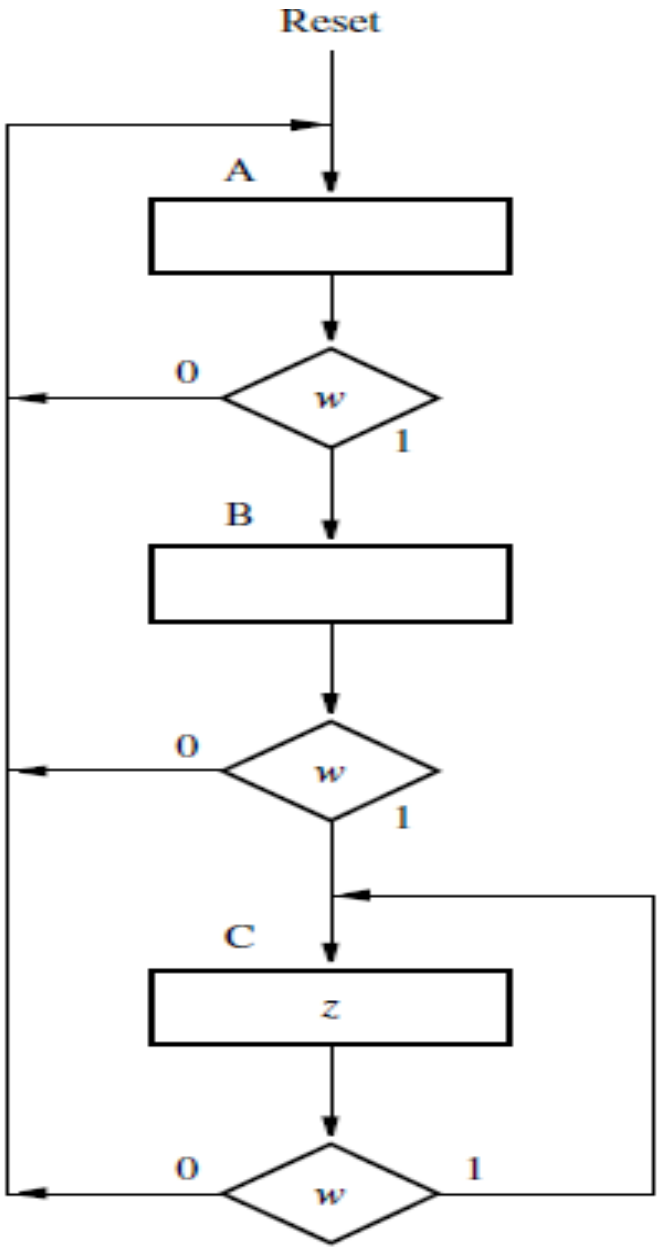


Figure 6.82 ASM chart for the FSM in Figure 6.3.

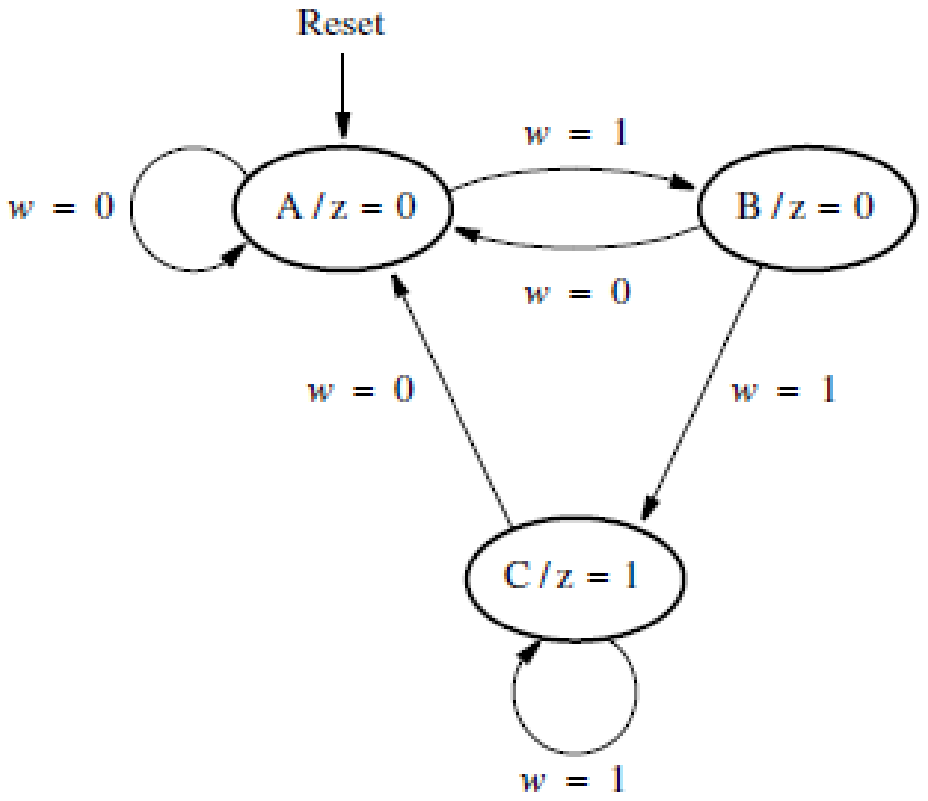


Figure 6.3 State diagram of a simple sequential circuit.

Mealy State Model

- output values are generated based on both the state of the circuit and the present values of its inputs.

Examlpe: with respect to the previous example, the output z should be equal to 1 in the same clock cycle when the second occurrence of $w = 1$ is detected.

Clock cycle:	t_0	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8	t_9	t_{10}
w:	0	1	0	1	1	0	1	1	1	0	1
z:	0	0	0	0	1	0	0	1	1	0	0

Figure 6.22 Sequences of input and output signals.

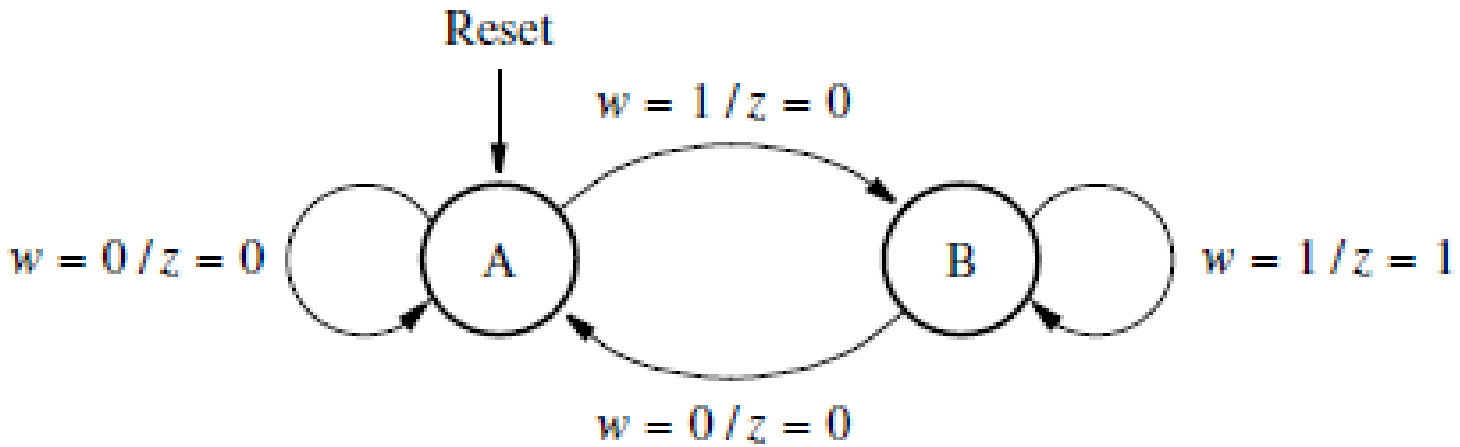


Figure 6.23 State diagram of an FSM that realizes the task in Figure 6.22.

Present state	Next state		Output z	
	$w = 0$	$w = 1$	$w = 0$	$w = 1$
A	A	B	0	0
B	A	B	0	1

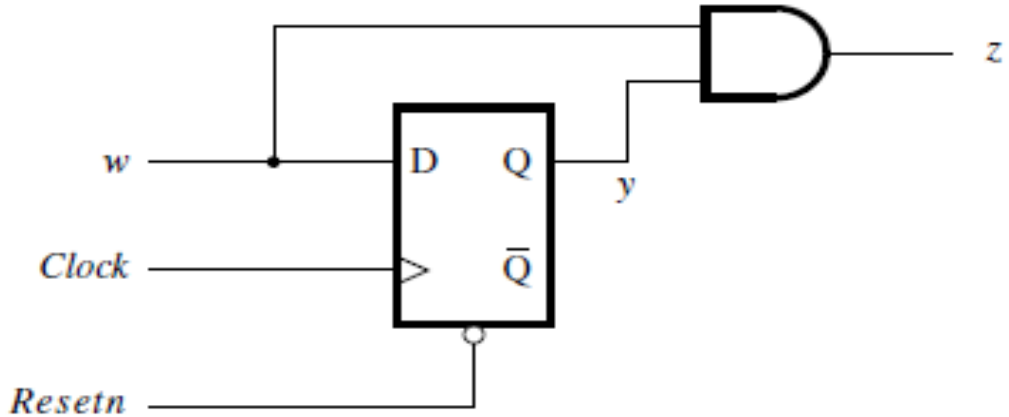
Figure 6.24 State table for the FSM in Figure 6.23.

	Present state	Next state		Output	
		$w = 0$	$w = 1$	$w = 0$	$w = 1$
	y	Y	Y	z	z
A	0	0	1	0	0
B	1	0	1	0	1

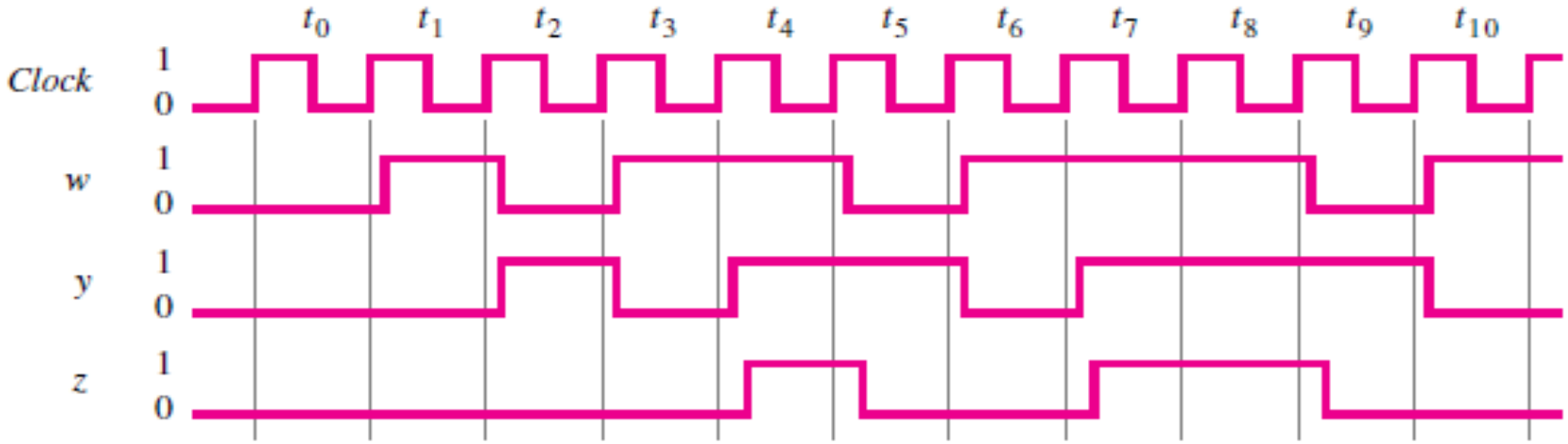
Figure 6.25 State-assigned table for the FSM in Figure 6.24.

$$Y = D = w$$

$$z = wy$$

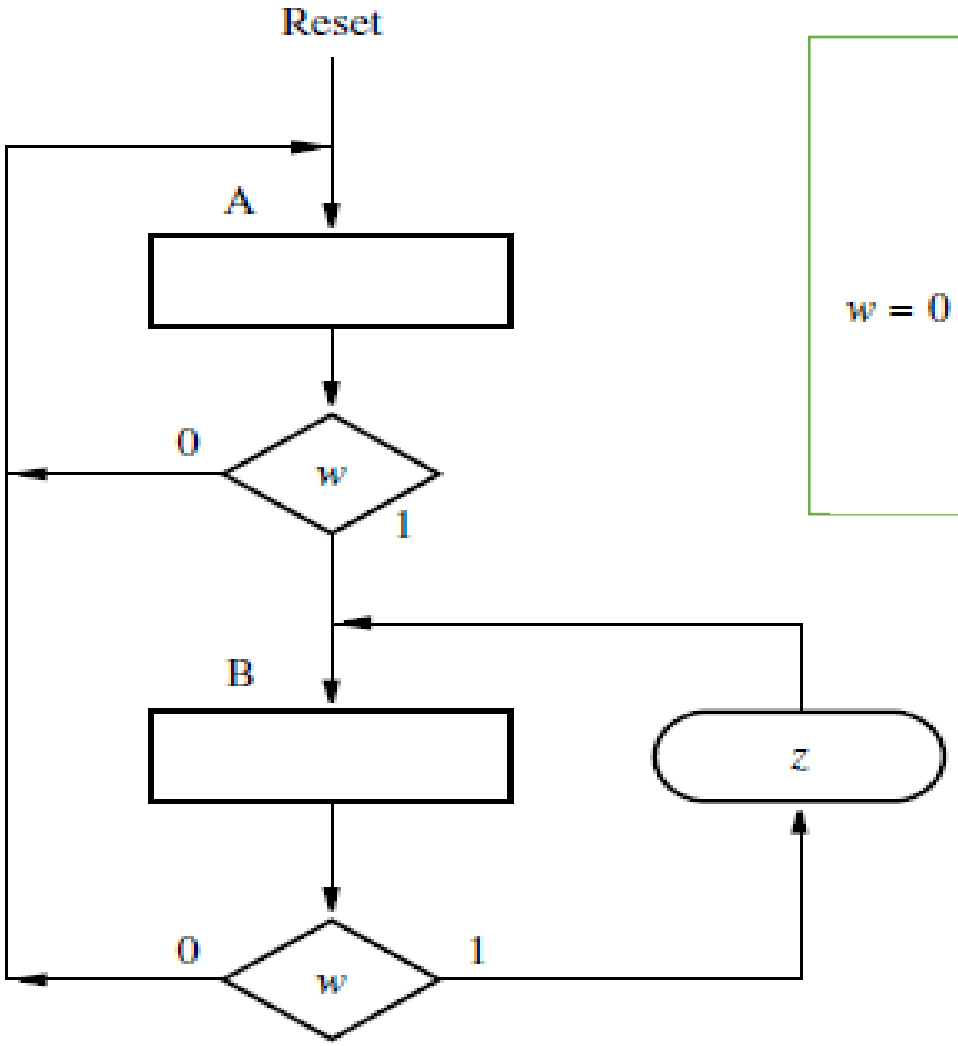


(a) Circuit



(b) Timing diagram

Figure 6.26 Implementation of the FSM in Figure 6.25.



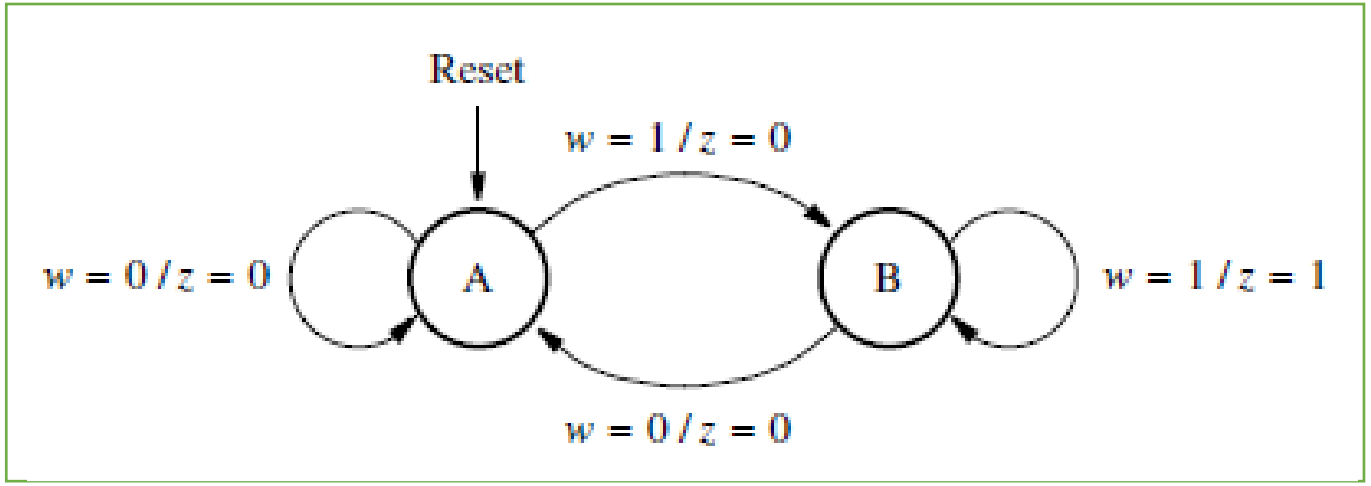


Figure 6.83 ASM chart for the FSM in Figure 6.23.

Example: swap the contents of two registers

R1←→R2
R3←R2
R2←R1
R1←R3
R3 is temp reg.

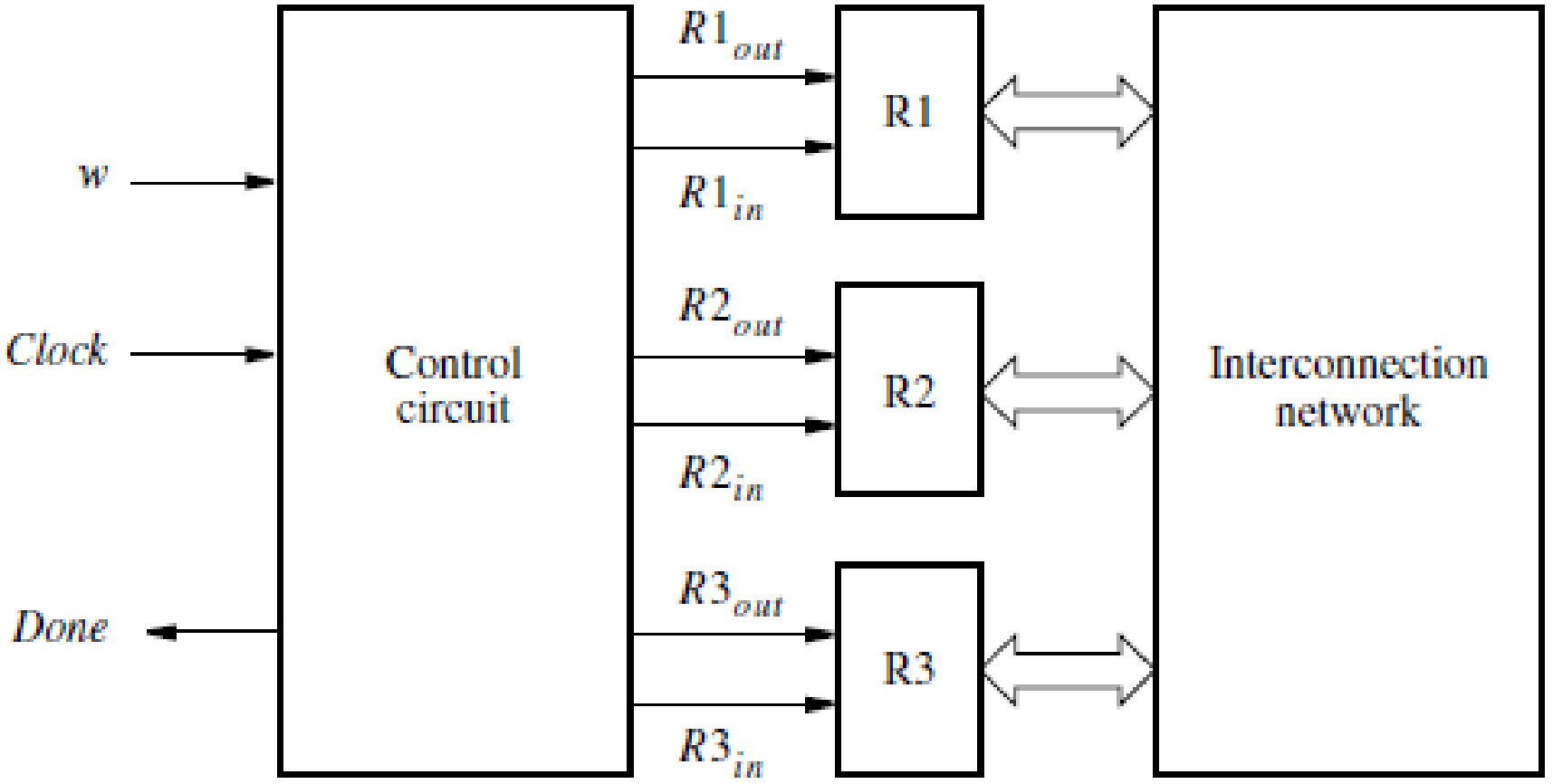


Figure 6.10 System for Example 6.1.

- Each register has two control signals: the *Rkout* signal causes the contents of register *Rk* to be placed into the interconnection network; the *Rkin* signal causes the data from the network to be loaded into *Rk*.
- The *Rkout* and *Rkin* signals are generated by a control circuit, which is a finite state machine.
- We will design a control circuit that swaps the contents of R1 and R2, in response to an input w = 1.
- Therefore, the inputs to the control circuit will be w and Clock. The outputs will be R1out , R1in, R2out , R2in, R3out , R3in, and Done which indicates the completion of the required transfers.

- The contents of $R2$ are first loaded into $R3$, using the control signals $R2out = 1$ and $R3in = 1$.
- Then the contents of $R1$ are transferred into $R2$, using $R1out = 1$ and $R2in = 1$.
- Finally, the contents of $R3$ (which are the previous contents of $R2$) are transferred into $R1$, using $R3out = 1$ and $R1in = 1$. Since this step completes the required swap, we will set the signal $Done = 1$.

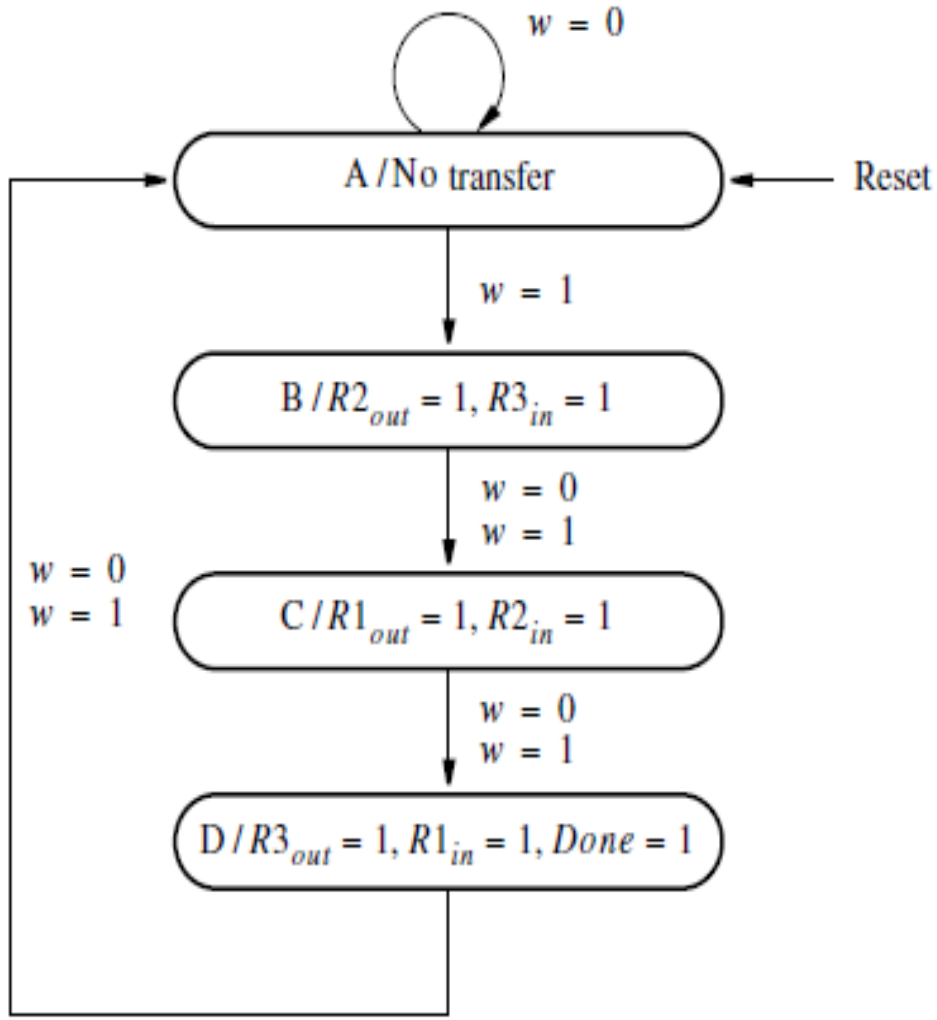


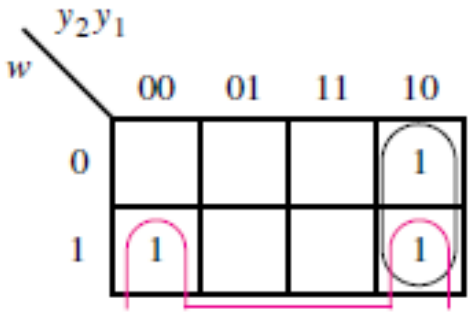
Figure 6.11 State diagram for Example 6.1.

Present state	Next state		Outputs						
	$w = 0$	$w = 1$	$R1_{out}$	$R1_{in}$	$R2_{out}$	$R2_{in}$	$R3_{out}$	$R3_{in}$	$Done$
A	A	B	0	0	0	0	0	0	0
B	C	C	0	0	1	0	0	1	0
C	D	D	1	0	0	1	0	0	0
D	A	A	0	1	0	0	1	0	1

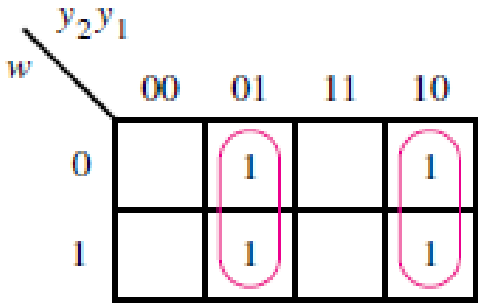
Figure 6.12 State table for Example 6.1.

Present state	Next state		Outputs						
	$w = 0$	$w = 1$	$R1_{out}$	$R1_{in}$	$R2_{out}$	$R2_{in}$	$R3_{out}$	$R3_{in}$	*Done*
A	A	B	0	0	0	0	0	0	0
B	C	C	0	0	1	0	0	1	0
C	D	D	1	0	0	1	0	0	0
D	A	A	0	1	0	0	1	0	1

	Present state	Next state		Outputs						
		$w = 0$	$w = 1$							
	y_2y_1	Y_2Y_1	Y_2Y_1	$R1_{out}$	$R1_{in}$	$R2_{out}$	$R2_{in}$	$R3_{out}$	$R3_{in}$	*Done*
A	0 0	0 0	0 1	0	0	0	0	0	0	0
B	0 1	1 0	1 0	0	0	1	0	0	1	0
C	1 0	1 1	1 1	1	0	0	1	0	0	0
D	1 1	0 0	0 0	0	1	0	0	1	0	1



$$Y_1 = w\bar{y}_1 + \bar{y}_1 y_2$$



$$Y_2 = y_1\bar{y}_2 + \bar{y}_1 y_2$$

The output control signals are derived as

$$R1_{out} = R2_{in} = \bar{y}_1 y_2$$
$$R1_{in} = R3_{out} = Done = y_1 y_2$$
$$R2_{out} = R3_{in} = y_1 \bar{y}_2$$